

HARYO BISMO WICAKSONO

Data Analytics | Forecasting | Business Intelligence | Machine Learning

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PROFESSIONAL SUMMARY

Final-year Data Science Technology student at Universitas Airlangga with hands-on experience in sales analytics, revenue forecasting, dashboard development, customer insight analysis, statistical modeling, and machine learning. Proficient in Python, SQL, R, Excel, Power BI, Tableau, and data visualization, with experience translating complex datasets into practical recommendations for business users. Proven leadership in cross-functional projects and international university collaboration, with strong interests in consumer, financial, and business analytics.

EDUCATION

B.Sc. Candidate, Data Science | Universitas Airlangga

Aug 2022 – Present

Surabaya, Indonesia

- GPA: 3.13/4.00 | Relevant coursework: Machine Learning, Statistical Analysis, Data Visualization, Database Systems, Natural Language Processing, and Mathematical Modeling.

PROFESSIONAL EXPERIENCE

Data Analyst Intern | Wisma Makara Universitas Indonesia

Jul 2025 – Aug 2025

Jakarta, Indonesia

- Analyzed historical hotel and facility sales data to identify revenue patterns, performance drivers, and business trends.
- Developed forecasting models for long-term revenue projections and six-month predictive analysis to support planning and decision-making.
- Built a management dashboard combining historical performance and forecast outputs, and communicated findings to non-technical stakeholders.

Part-Time Social Media Administrator | KlinikKeuangan.com

2018 – 2022

Remote

- Managed digital content, moderated audience discussions, and compiled social media interaction and performance reports for a personal financial planning platform.

PROJECTS

Sentiment Analysis of Maybank Indonesia Branch Reviews

May 2025

Customer Insight & NLP Project

- Analyzed 1,737 Google Reviews across Maybank Indonesia branches using text preprocessing, MiniLM embeddings, sentiment classification, topic modeling, and geographic visualization.
- Compared Logistic Regression, Random Forest, and CatBoost; Logistic Regression achieved the highest F1-score of 88.7%.
- Identified key customer concerns related to service speed, queues, staff responsiveness, and branch facilities, then translated the findings into actionable service-improvement recommendations.

Macroeconomic Analysis and LQ45 Forecasting Using VECM and LSTM

May 2026 – Jul 2026

Undergraduate Thesis

- Analyzed the relationship between the LQ45 stock index and USD/IDR exchange rates, inflation, and Bank Indonesia's policy rate using daily data from 2010 to 2025.
- Developed VECM and LSTM pipelines covering stationarity testing, Johansen cointegration, impulse response analysis, hyperparameter tuning, and recursive five-day forecasting.
- Achieved RMSE of 10.56, MAE of 7.95, MAPE of 0.94%, and R^2 of 0.983 with the tuned LSTM model on the test set.
- Generated complementary insights by using VECM for short- and long-term economic interpretation and LSTM for nonlinear forecasting.

Indonesian News Text Summarization Using Hybrid T5 and

Oct 2024 – Dec 2024

LexRank

Natural Language Processing Project

- Compared nine extractive, abstractive, and hybrid summarization approaches on 100 Indonesian news articles with manually annotated 5W+1H reference summaries.
- Evaluated model performance using ROUGE metrics and qualitative assessment of relevance, coherence, completeness, and information preservation.
- Achieved the strongest performance with the hybrid T5 + LexRank model, recording ROUGE-1 of 0.526, ROUGE-2 of 0.430, and ROUGE-L of 0.497.
- Demonstrated that combining LexRank's sentence-selection capability with T5's semantic generation improved summary coverage and readability.

Statistical Modeling of C-Class Solar Flare Counts

May 2026 – Jul 2026

Categorical Data Analysis Project

- Modeled 1,389 active-region observations using Poisson, Generalized Poisson, and Negative Binomial regression, including MLE, Newton-Raphson estimation, overdispersion tests, and IRR interpretation.
- Performed maximum likelihood estimation, Newton-Raphson optimization, overdispersion testing, model diagnostics, likelihood-ratio testing, and Incidence Rate Ratio interpretation.
- Identified Zurich class, spot size, activity, and largest-spot area as important predictors, with Zurich classes E and F associated with higher flare counts.

- Selected Negative Binomial regression based on the lowest AIC and significant likelihood-ratio improvement, then developed a flare-risk classification framework.

VOLUNTEER & ORGANIZATIONAL EXPERIENCE

Directorate General, Global Network Collaboration | BEM FTMM

Feb 2023 – Dec 2024

Surabaya, Indonesia

- Being a member of the Ministry of External Affairs at BEM FTMM, accountable for fostering international relationships to facilitate global collaborations and presence via strategic networking and outreach initiatives
- Led international outreach and collaboration initiatives across two organizational terms and supported institutional visits involving Universiti Teknologi Petronas and Universiti Malaya.
- Spearhead efforts to establish international partnerships for BEM FTMM, negotiate collaboration agreements with overseas universities, drive initiatives to promote cooperation and knowledge exchange for mutual benefit. Part of the organizing committee and welcoming team for Universiti Teknologi Petronas visitation team in 2023.
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Lead Project Officer & Student Outbound Delegate | FTMM FADE-IN × UTM

Aug 2024 – Nov 2024

Surabaya & Johor Bahru

- Led preparation and on-site execution of an international student outbound program at Universiti Teknologi Malaysia, coordinating cross-functional teams and university stakeholders.
- One of the 20 students selected to represent FTMM Universitas Airlangga in academic and cross-cultural engagement activities with Universiti Teknologi Malaya in Johor Bahru.

Head of Event Division | KARSA FTMM 2023

May 2023 – Aug 2023

Surabaya, Indonesia

- Managed schedules, resources, event operations, and cross-divisional coordination to ensure timely delivery of program milestones.
- Led the processes to deliver efficient execution of tasks, spearheaded the preparation, organization, and overall coordination of the event, overseeing every detail of the event to ensure successful planning and implementation, while managing resources to meet project goals.

Lead Project Officer | ScholPrep Scholarship Preparation Talk Show

May 2023 – Nov 2023

Surabaya, Indonesia

- Coordinated cross-divisional planning and execution, maintained project timelines, and ensured smooth communication across teams.

Public Relations & Sponsorship Team | Airlangga Talks 6.0

Apr 2023 – May 2023

Surabaya, Indonesia

- Conducted speaker and sponsor outreach, supported social media communications, and contributed to partnerships with organizations including Danone and Telkomsel.

Co-Founder & Chairman | FORDESI Student Debate Forum

Jun 2021 – Dec 2021

Jakarta, Indonesia

- Co-founded and led a student debate forum, delivering training and simulations for parliamentary debate competitions.

SKILLS

Programming & Data: Python, SQL, R, VBA, Pandas, NumPy, Scikit-learn, TensorFlow, Statsmodels, SPSS, Minitab

Analytics & Modeling: Data analysis, business analytics, forecasting, time-series analysis, machine learning, NLP, statistical modeling, customer insight analysis

Visualization & Tools: Microsoft Excel, Power BI, Tableau, Matplotlib, dashboard development, Jupyter Notebook

Professional: Analytical thinking, project management, stakeholder communication, cross-functional collaboration, cross-cultural communication

Languages: Indonesian, English, French (DELF A2)